Bivariate data with technology

1

a $y = -0.29x + 0.67$

b $y = 0.12x - 1.75$

2

a

i -0.47

ii Weak negative linear correlation

b

i -0.84

ii Strong negative linear correlation

c

i 0.47

ii Weak positive linear correlation

d

i -0.91

ii Strong negative linear correlation

e

i -0.59

ii Moderate negative linear correlation

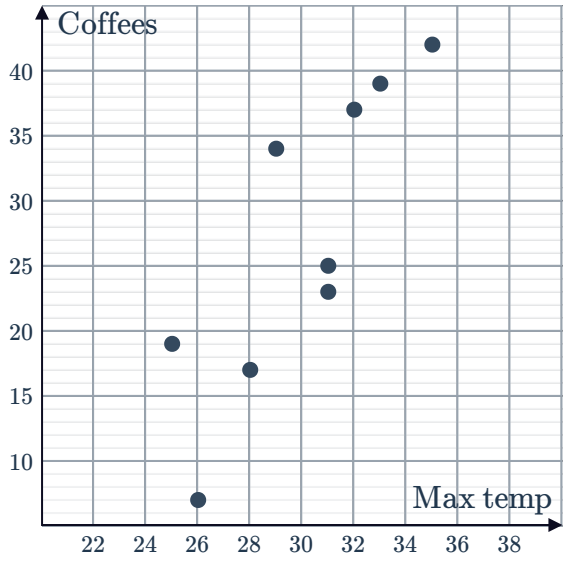
f

i 0.19

ii Weak positive linear correlation

3

a



b $r = 0.84$

c As the temperature increases, the sales tend to increase.

4

a -0.85

b Strong negative linear relationship

c $y = -0.1x + 30.3$

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5

a 0.60

b Moderate positive linear relationship

c $y = 4.2x - 34.5$

6

a $y = -1694x + 18\,407$

b $18\,407$

c $\$18\,407$

d -1694

e The gradient of the line indicates that age and price have a negative correlation.

f The price decreases by $\$1694$.

Predictions

7

a $r = -0.908$

b Strong negative linear relationship

c $y = -0.06x + 6.37$

d $6.07\ \text{kg}$

e Reliable due to interpolation and a strong correlation.

8

a $r = 0.997$

b Strong positive linear relationship

c $y = 0.85x + 1.28$

d 52.28%

e Reliable due to interpolation and a strong correlation.

9

a $V = 14.9P + 30.3$

b 30.3

c 30.3 words

d Yes, as children learn words in other ways.

e 14.9

f For each additional page that a child reads each day, the child's vocabulary will increase by 14.9 words.

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10

a $P = -0.2M + 99.6$

b 99.6

c 99.6%

d Yes, it is reasonable because they will have more time to study.

e -0.2

f For each additional minute that a student uses SnappyChatty, their test mark will decrease by 0.2%.

11

a $r = 0.79$

b Strong positive linear relationship

c $y = 0.025x - 550.108$

d $\$1449.89$

e $\$80\,000$ because the other two are examples of extrapolation.

12

a $r = 0.86$

b Strong positive linear relationship

c $y = 0.89x + 2.78$

d $25.03^\circ C$

e 32°C, because the other two values are examples of extrapolation.

13

a $r = -0.91$

b Strong negative linear relationship

c $y = -6.6x + 80.1$

d 47.1

e Reliable due to interpolation and a strong correlation.

14

a 0.72

b Strong positive linear relationship

c $A = 1.61E - 16.63$

d 39.72%

e **C**

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